B.Comp. Dissertation

DQN-Guided GRPO Fine-tuning of Traffic Signal Control Agent

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School of Computing
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Abstract

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1 Introduction

Rapid urbanization and accelerating motorization have placed unprecedented strain on urban transportation systems worldwide. As cities densify, limited road infrastructure must accommodate ever-growing travel demand, leading to persistent congestion, unreliable travel times, and increased environmental externalities. Inefficient traffic flow not only wastes fuel and economic productivity but also amplifies air pollution and degrades quality of life, particularly in densely populated urban corridors.

Traffic signal control (TSC) aims to coordinate phase sequences and durations at signalized intersections to maximize throughput and reduce congestion. Classical fixed-time methods use the Webster formula[17] to preset signal timings based on historical traffic data collected from the traffic network at different times. In contrast, the max-pressure control framework[7] computes a pressure value for each signal phase from the difference between queue lengths on opposing approaches and selects the phase with the highest pressure; this strategy can be shown to maximize network throughput.

Reinforcement-learning (RL) methods model TSC as a Markov decision process in which an agent observes traffic states (for example, lane queue lengths and delays), chooses an action (phase switch) and receives a reward linked to throughput or delay reduction. Deep RL further parameterizes the control policy using neural networks, enabling the agent to approximate complex, high-dimensional decision functions. Such methods have demonstrated strong empirical performance across a wide range of simulated traffic scenarios. However, they also introduce notable challenges. In particular, learned policies often exhibit limited generalization, as models trained in one traffic environment exhibit degraded performance when deployed in different cities or under unseen conditions. Furthermore, the reliance on opaque neural representations reduces interpretability, making it difficult to explain or verify decision-making processes. This lack of transparency poses a barrier to real-world adoption, where reliability and accountability are critical.

Large Language Models (LLMs) represent a new paradigm in artificial intelligence, combining strong reasoning ability with the capacity to generate interpretable, text-based outputs. These properties make LLMs promising candidates for traffic signal control, where decision transparency and adaptability are highly desirable. By framing control as a reasoning process, LLM-based agents can potentially provide both actionable policies and human-understandable explanations, bridging the gap between performance and interpretability. However, this potential is not yet fully realized in practice. Achieving real-time responsiveness typically requires deploying smaller-scale models or mixture-of-experts (MoE) architectures with compact expert networks. Without sufficient domain adaptation, such models often fail to capture complex traffic dynamics, leading to suboptimal decisions.

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2 Related Work

Traffic Signal Control. Early TSC methods such as *FixedTime* rely on predefined cycle lengths and manually designed phase splits for each intersection. Although these

approaches are simple and stable in practice, they adapt poorly to time-varying traffic demand. *MaxPressure*[7] improves adaptability by selecting phases according to the queue imbalance between incoming and outgoing directions, making it more responsive than fixed plans. However, both *FixedTime* and *MaxPressure* remain constrained by handcrafted rules or preset logic, which limits their flexibility under complex and highly dynamic traffic scenarios.

Recent progress in reinforcement learning has led to a series of RL-based TSC methods. *CoLight*[16] models network-level cooperation among intersections through graph attention, enabling each signal to coordinate with surrounding intersections instead of acting independently. *CoSLight*[10] further extends this idea by jointly optimizing signal control and collaborator selection, so that each intersection can dynamically decide which neighboring intersections are most useful for coordination under current traffic conditions. *MPLight*[1] is a decentralized and pressure-aware RL approach designed for large-scale road networks, using parameter sharing to improve scalability and training efficiency. Although these methods achieve strong performance in per city simulation, RL-based controllers face challenges in generalization and interpretability, hence deployment robustness in real-world environments.

To address these limitations, recent studies have begun exploring the integration of large language models (LLMs) into TSC. *LLM-Assisted Light*[15] places an LLM at the center of control and augments it with perception and decision-support tools so that it can combine live traffic observations with established TSC methods and remain robust under emergency, roadblock, and sensor-outage scenarios. *iLLM-TSC*[9] instead adopts a hybrid design in which an RL controller first proposes an action and an LLM then verifies or adjusts that action when observations are degraded or rare events such as packet loss and emergency vehicles fall outside the original RL formulation. *LATS*[18] further explores a teacher-student setting in which an LLM teacher provides rich semantic features during multi-agent RL training, while a smaller student controller learns to emulate those features and is deployed without the LLM at inference time. *LLMLight*[5] moves closer to end-to-end LLM control by adapting its LightGPT backbone through imitation fine-tuning and critic-guided policy refinement so that the model can make direct signal decisions with improved interpretability and cross-scenario performance. *Traffic-R1*[19] extends this trend with a 3B RL-finetuned foundation model that combines expert-aligned offline training, online self-exploration, and asynchronous multi-agent communication to achieve strong zero-shot and out-of-distribution TSC performance. Compared with end-to-end LLM finetuning, LLM+RL hybrids may remain inferior because they still inherit the ceiling and failure modes of the underlying RL controller. Existing end-to-end finetuning methods also have notable limitations: In *LLMLight*, the critic-guided policy refinement objective is weaker than modern policy-gradient methods because the reward/advantage magnitude does not directly participate in loss computation. *Traffic-R1*, meanwhile, relies on substantial human effort to label the offline finetuning data, and its fixed-horizon cumulative queue-based reward assigns the same coarse mean reward to all steps within the horizon.

RL finetuning of LLMs. Large language models are commonly adapted to downstream tasks through instruction finetuning or preference alignment, and instruction finetuning has become a popular baseline because it is simple, stable, and effective at teaching a model to follow task-specific formats and instructions [2, 8]. By contrast, RL-based finetuning has traditionally been less common in downstream applications because RLHF-

style pipelines usually require supervised warm-starting, reward or preference modeling, online rollout collection, and carefully stabilized policy optimization, all of which increase engineering complexity and computational cost [8, 13]. Recent progress nevertheless shows that reinforcement learning can elicit reasoning capabilities that are difficult to obtain by supervised imitation alone: *DeepSeekMath* introduced Group Relative Policy Optimization (GRPO), which removes the separate value network and replaces it with group-relative advantages to reduce memory overhead while retaining PPO-style policy updates [12], and *DeepSeek-R1* further demonstrated that large-scale self-exploration can substantially improve reasoning behavior in LLMs [4]. These developments make RL fine-tuning increasingly attractive for domain-specific agents, since it can reduce dependence on large manually curated datasets, optimize directly for long-horizon behavioral objectives instead of next-token imitation, and preserve useful pretrained abilities through KL-constrained updates around a reference policy [8, 12, 4].

3 Preliminaries

3.0.1 Reinforcement Learning Overview

Reinforcement learning (RL) models sequential decision making as a Markov decision process (MDP) $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$. At each time step t, the agent observes a state $s_t \in \mathcal{S}$, selects an action $a_t \in \mathcal{A}$ according to a policy $\pi_\theta(a_t \mid s_t)$, receives a scalar reward $r_t = r(s_t, a_t)$, and the environment transitions to $s_{t+1} \sim p(\cdot \mid s_t, a_t)$. The objective is to find policy parameters θ that maximize the expected discounted return

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{\infty} \gamma^t r_t\right],$$

where $\gamma \in (0, 1]$ is the discount factor. For a fixed policy π, the state-value function and action-value function are

$$V^\pi(s) = \mathbb{E}_\pi\big[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s\big], \qquad Q^\pi(s, a) = \mathbb{E}_\pi\big[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s, a_0 = a\big],$$

and they satisfy the Bellman expectation equation

$$V^\pi(s) = \sum_a \pi(a \mid s) \sum_{s'} p(s' \mid s, a)\big[r(s, a) + \gamma V^\pi(s')\big].$$

The advantage function is defined as

$$A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s),$$

and measures how much better action a is compared to the average action at state s.

Policy gradient principle. For differentiable policies $\pi_\theta(a \mid s)$, the policy gradient theorem gives

$$\nabla_\theta J(\theta) = \mathbb{E}_{s \sim d^\pi, a \sim \pi_\theta}\big[\nabla_\theta \log \pi_\theta(a \mid s)\, Q^\pi(s, a)\big],$$

and in practice we replace $Q^\pi(s,a)$ by an advantage estimate $\hat{A}_t$, leading to the classic REINFORCE-style update

$$\theta \leftarrow \theta + \alpha\, \hat{A}_t\, \nabla_\theta \log \pi_\theta(a_t \mid s_t).$$

Most modern RL methods (PPO, GRPO, and their variants) can be viewed as safer or more sample-efficient ways of performing this update.

Proximal Policy Optimization (PPO). PPO [11] stabilizes policy gradients by preventing excessively large policy updates. Let

$$r_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}$$

be the probability ratio between the new and the behavior policy. Let $\hat{A}_t$ be an advantage estimate (e.g., GAE). PPO maximizes the clipped surrogate objective

$$L^{\text{PPO}}(\theta) = \mathbb{E}_t\Big[\min\big(r_t(\theta)\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\big)\Big],$$

where ϵ is a small hyperparameter (e.g., 0.1 or 0.2). The $\min(\cdot)$ and the clipping together act as a trust region in probability space, so that even if $\hat{A}_t$ is large, the policy will not move too far away from $\pi_{\theta_{\text{old}}}$ in a single update.

Group Relative Policy Optimization (GRPO). Recent LLM-oriented RL work observes that when many responses can be sampled for the same prompt, we can estimate advantages without a separate value network. GRPO forms a group $\mathcal{G} = \{r_1, \ldots, r_N\}$ of rollouts for the same input, scores them with a reward model $R_\phi(\cdot)$, and then normalizes within the group:

$$A_i = \frac{R_\phi(r_i) - \text{mean}(\mathcal{G})}{\text{std}(\mathcal{G})}.$$

This gives a critic-free advantage estimate. GRPO then uses essentially the same clipped PPO-style loss

$$L^{\text{GRPO}}(\theta) = \frac{1}{N}\sum_{i=1}^{N} \min\left(\frac{\pi_\theta(r_i \mid x)}{\pi_{\theta_{\text{old}}}(r_i \mid x)}A_i, \text{clip}\left(\frac{\pi_\theta(r_i \mid x)}{\pi_{\theta_{\text{old}}}(r_i \mid x)}, 1-\epsilon, 1+\epsilon\right)A_i\right) - \lambda\, D_{\text{KL}}\big(\pi_\theta \,\|\, \pi_{\text{ref}}\big),$$

where the KL term keeps the updated policy close to a reference model (often the SFT model). Compared to PPO, the key differences are: (i) advantages come from groupwise normalization, not a learned value function; (ii) the method is well suited to large-batch LLM training where many samples per prompt are available.

Group-in-Group Policy Optimization (GiGPO). Very recent work on training LLM agents in multi-step environments argues that a single group-level relative advantage (as in GRPO) is not enough, because the reward can be sparse and credit assignment needs to happen both at the trajectory level and at individual steps. GiGPO [3] introduces a two-level grouping:

- at the episode level, sample multiple full trajectories for the same task and compute a macro relative advantage (similar to GRPO);
- at the step level, retroactively group actions that occurred in the same environment state (called anchor states) across different trajectories, and compute a micro relative advantage for those actions.

This yields fine-grained credit signals without introducing a value network. Conceptually, GiGPO keeps the good parts of GRPO (critic-free, low memory, PPO-like clipping) but adds an extra axis of grouping so that actions that happened at the same state can be compared directly. This is particularly attractive for long-horizon urban/traffic tasks or LLM agents, where the effect of a single bad action might only become visible many steps later.

3.0.2 Traffic Signal Control

General overview. Traffic signal control (TSC) aims to coordinate phase sequences and durations at signalized intersections to maximize throughput and reduce congestion. Classical fixed-time methods use the Webster formula[17] to preset signal timings based on historical traffic data collected from the traffic network at different times. In contrast, the max-pressure control framework[7] computes a pressure value for each signal phase from the difference between queue lengths on opposing approaches and selects the phase with the highest pressure; this strategy can be shown to maximize network throughput. Reinforcement-learning (RL) methods model TSC as a Markov decision process in which an agent observes traffic states (for example, lane queue lengths and delays), chooses an action (phase switch) and receives a reward linked to throughput or delay reduction. CoLight[16] learns network-level cooperation by letting intersections communicate through graph attention, so signals can coordinate beyond immediate neighbors. CoSLight[10] extends this idea by learning not only the signal policy but also which intersections should collaborate at each moment, improving multi-intersection coordination under changing congestion patterns. MPLight[1] is a decentralized, pressure-aware RL variant built to run efficiently in large CityFlow[14]-style networks, using parameter sharing to keep city-scale control tractable. Despite achieving competitive performance in simulation, many RL-based controllers suffer from poor cross-city generalization and lack interpretability, motivating recent research into interpretable and generalizable TSC agents.

CityFlow environment. A key enabler for modern RL research on TSC is the availability of scalable simulators. SUMO[6], the traditional open-source traffic simulator, is often too slow for city-scale RL experiments. CityFlow[14] is a multi-agent microscopic traffic simulator designed for large-scale RL studies. It provides flexible definitions of road networks and traffic flows from both synthetic and real-world data and exposes a user-friendly API for interaction with RL agents. CityFlow's multithreaded design makes it more than twenty times faster than SUMO, enabling efficient simulation of networks with hundreds or thousands of intersections. Because of its speed and extensibility, CityFlow has become a standard environment for evaluating RL-based TSC algorithms.

***LLMLight*: Large language models as TSC agents.** The *LLMLight*[5] framework treats traffic signal control as a partially observable Markov game and leverages large language models (LLMs) to serve as decision-making agents. At each signal switch, an agent collects observation features such as the number of queuing vehicles and approaching vehicles on each incoming lane, augments this information with a traffic scenario description, a task description and a commonsense knowledge prompt that encourages the agent to prioritize long queues, and feeds the combined text to an LLM. The LLM performs chain-of-thought reasoning to decide which phase to activate and provides a human-readable rationale. *LLMLight* constructs a specialized backbone model called LightGPT via imitation fine-tuning on GPT-4 trajectories and critic-guided policy refinement. During imitation, the model minimizes the negative log-likelihood

$$\mathcal{L}_{\text{NLL}} = -\sum_{t=1}^{T} \log p_\theta\big(y_t \mid y_{<t}, \pi, \mathcal{P}\big),$$

where π is the prompt consisting of observation and commonsense instructions and $\mathcal{P}$ denotes the reasoning trajectory. A critic model derived from an action-value network evaluates sampled trajectories, and a ranking-based loss steers the model toward trajectories with higher long-term rewards. Experimental results show that *LLMLight* with LightGPT achieves competitive or superior performance and provides better interpretability compared with traditional RL baselines across diverse traffic scenarios. It provides a reproducible workflow that integrates CityFlow with RL-based fine-tuning of LLMs (though implemented in TensorFlow) and datasets ready for use.

***Traffic-R1*: Reinforced LLMs for human-like TSC.** *Traffic-R1*[19] is a 3B foundation model trained to perform traffic signal control with human-like reasoning. It adopts a two-stage reinforcement-learning fine-tuning paradigm: an offline stage aligns the model with question-answer (QA) samples (generated by DeepSeek-R1 and verified by human experts), and an online stage allows self-exploration in simulation to refine the policy via reward feedback. The reward is calculated from the cumulative queuing time after multiple steps, and each step only half of the agents are activated, whose decisions are propagated to neighboring agents as an input. This mimics an asynchronous communication network allows multiple agents to coordinate across intersections. Benchmark results report that *Traffic-R1* offers zero-shot generalization to unseen road networks and out-of-distribution incidents, outperforming both classical heuristics and RL controllers. When deployed in production, *Traffic-R1* manages signals affecting more than 55,000 drivers each day, reducing average queue lengths by over 5% and significantly lowering operator workload. Though without a codebase, *Traffic-R1* provides insights into using SOTA RL algorithms like GRPO to train TSC agents, as well as the innovative idea of mimicking the real world scenario with a asynchronous mult-agent setup.

4 Plan for Next Semester

The plan for the next semester is outlined as follows:

1. Refactor the existing TensorFlow-based codebase to PyTorch for improved com-

patibility and flexibility, and reorganize the current project structure for better maintainability.

2. Adapt the GiGPO-style reward calculation to the TSC context. Test and make necessary adjustments to the formulation and give it a cool name.

3. Implement the asynchronous message propagation mechanism as described in the *Traffic-R1* paper.

4. Conduct comprehensive experiments and benchmark the proposed framework against state-of-the-art (SOTA) baselines.

5. Prepare and finalize the report documenting the methodology, results, and findings.

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